

TCA Cycle

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TCA Cycle

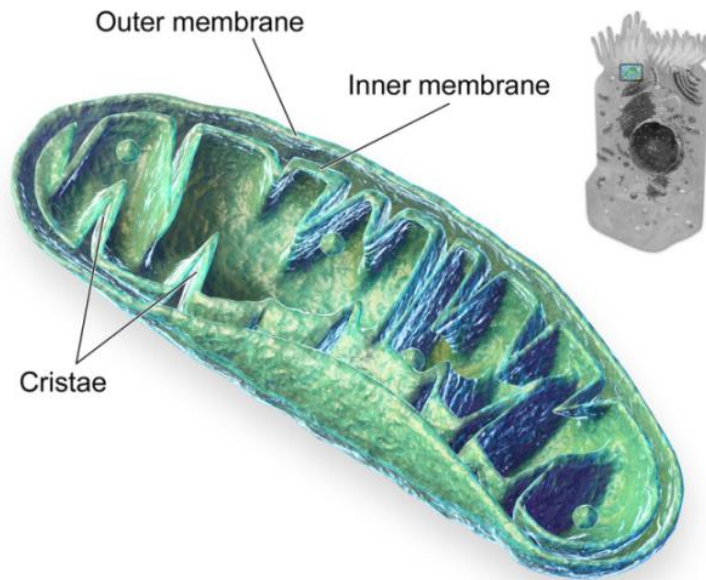
Tricarboxylic Acid Cycle, Krebs Cycle, Citric Acid Cycle

- Metabolic pathway
- Converts acetyl-CoA $\rightarrow$ CO₂
- Derives energy from reactions

TCA Cycle

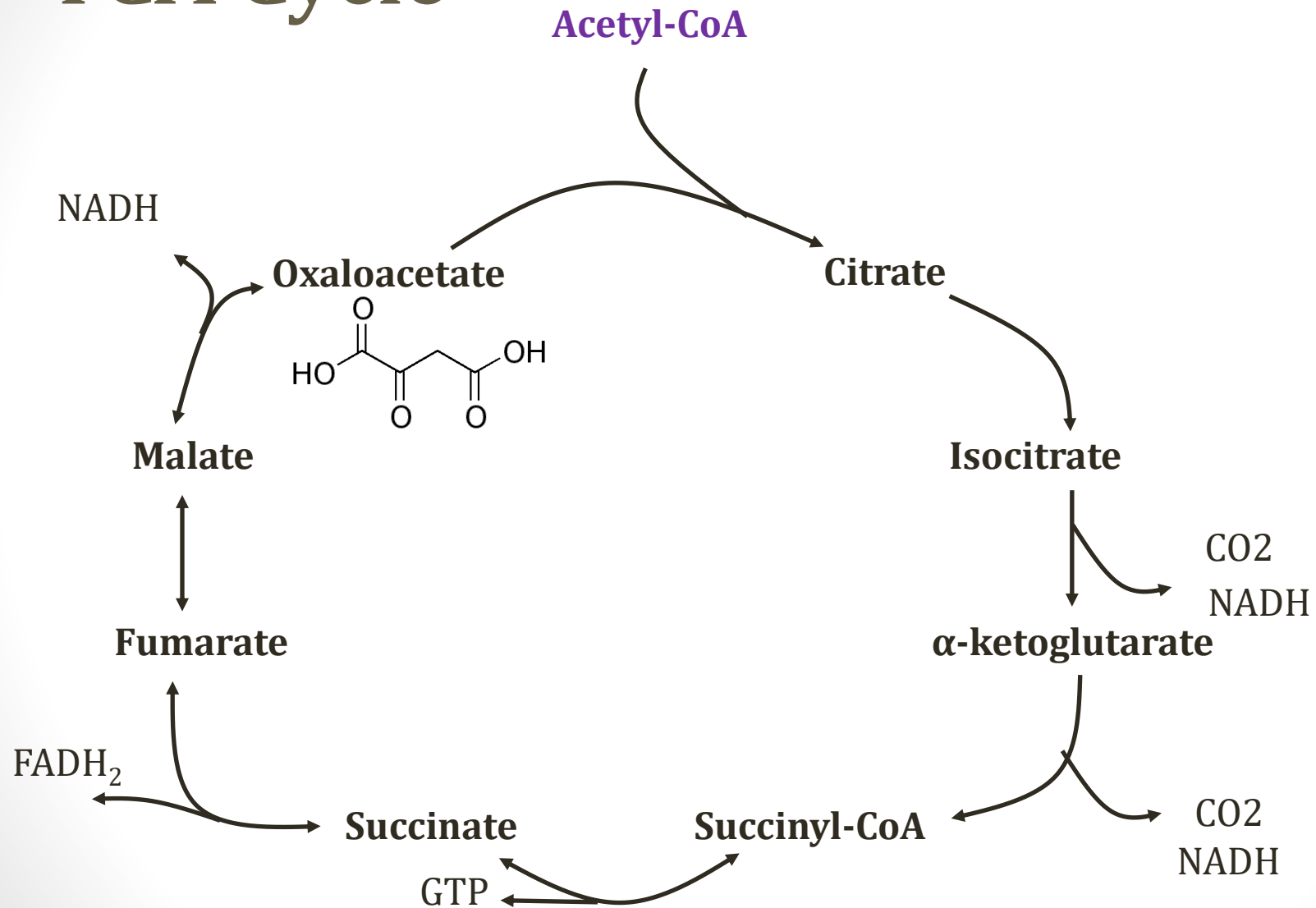
Tricarboxylic Acid Cycle, Krebs Cycle, Citric Acid Cycle

- All reactions occur in mitochondria
- Produces:
 - NADH, $\text{FADH}_2 \rightarrow$ electron transport chain (ATP)
 - GTP
 - CO_2



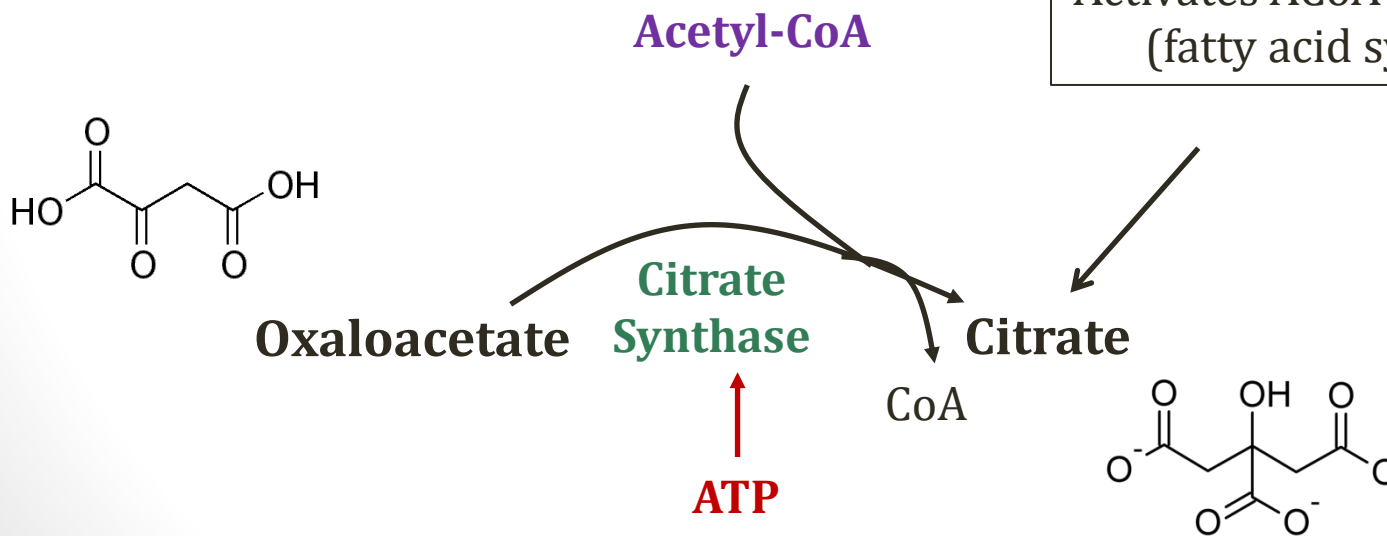
Mitochondria

TCA Cycle



Citrate Synthesis

- 6 Carbon structure
- Oxaloacetate (4C) + Acetyl-CoA (2C)
- Inhibited by **ATP**

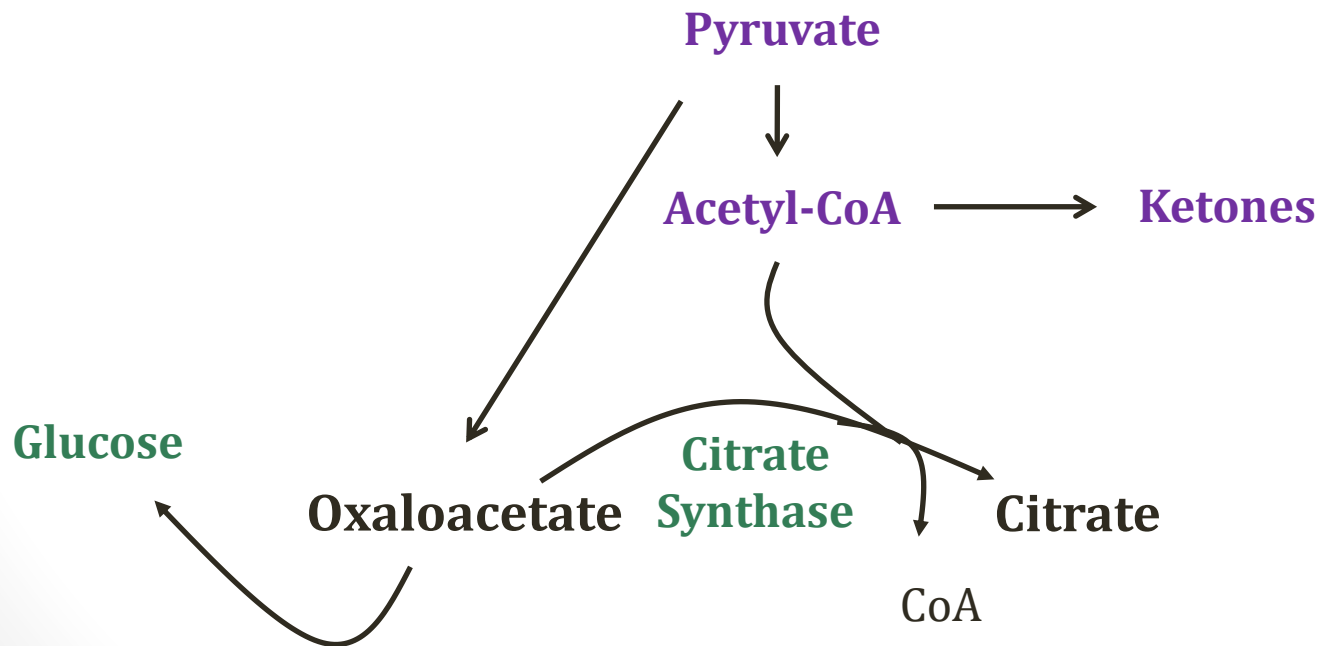


Special Points:

Inhibits PFK1 (glycolysis)
Activates ACoA carboxylase
(fatty acid synthesis)

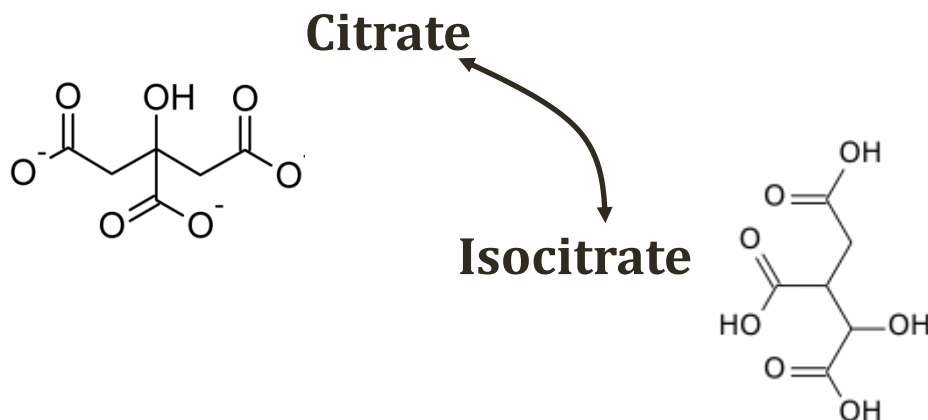
Fasting State

- Oxaloacetate used for gluconeogenesis
- ↓ **oxaloacetate** for TCA cycle
- Acetyl-CoA (fatty acids) → Ketone bodies



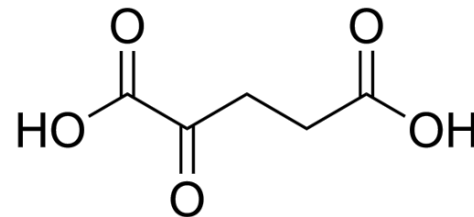
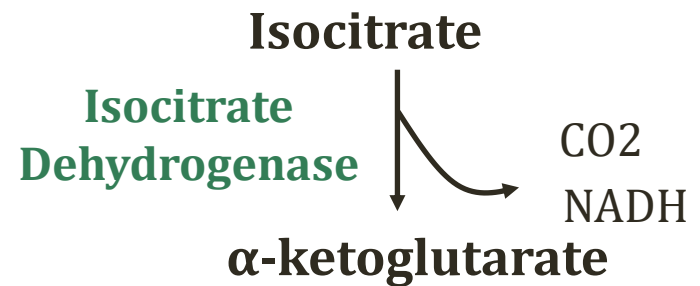
Isocitrate

- Isomer of citrate
- Enzyme: aconitase
- Forms intermediate (cis-aconitate) then isocitrate
- Inhibited by fluoroacetate: rat poison



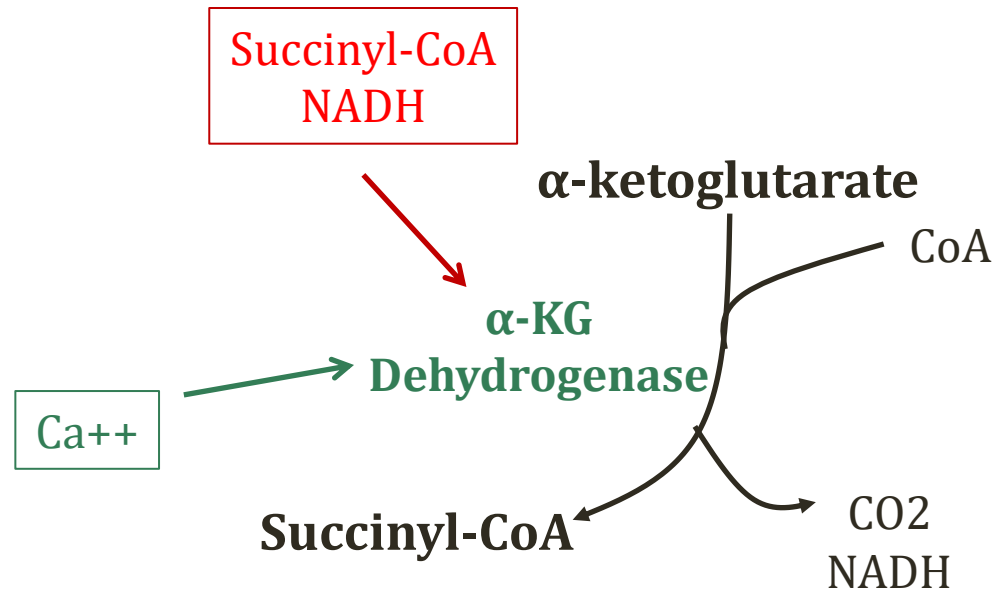
α -Ketoglutarate

- Rate limiting step of TCA cycle
- Inhibited by:
 - ATP
 - NADH
- Activated by:
 - ADP
 - Ca^{++}



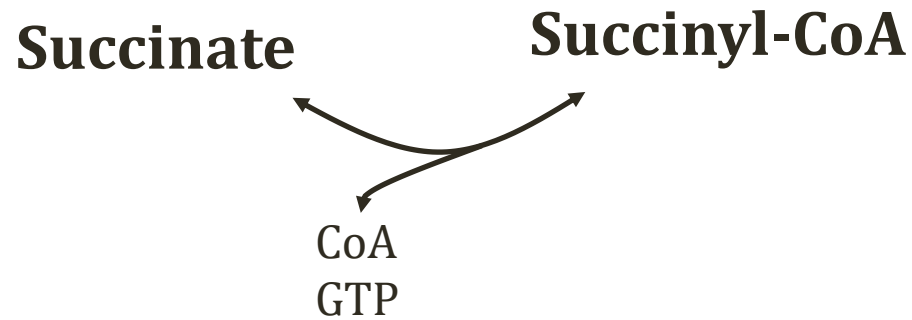
Succinyl-CoA

- α -ketoglutarate dehydrogenase complex
- Similar to pyruvate dehydrogenase complex
- Cofactors:
 - Thiamine
 - CoA
 - NAD
 - FADH
 - Lipoic acid



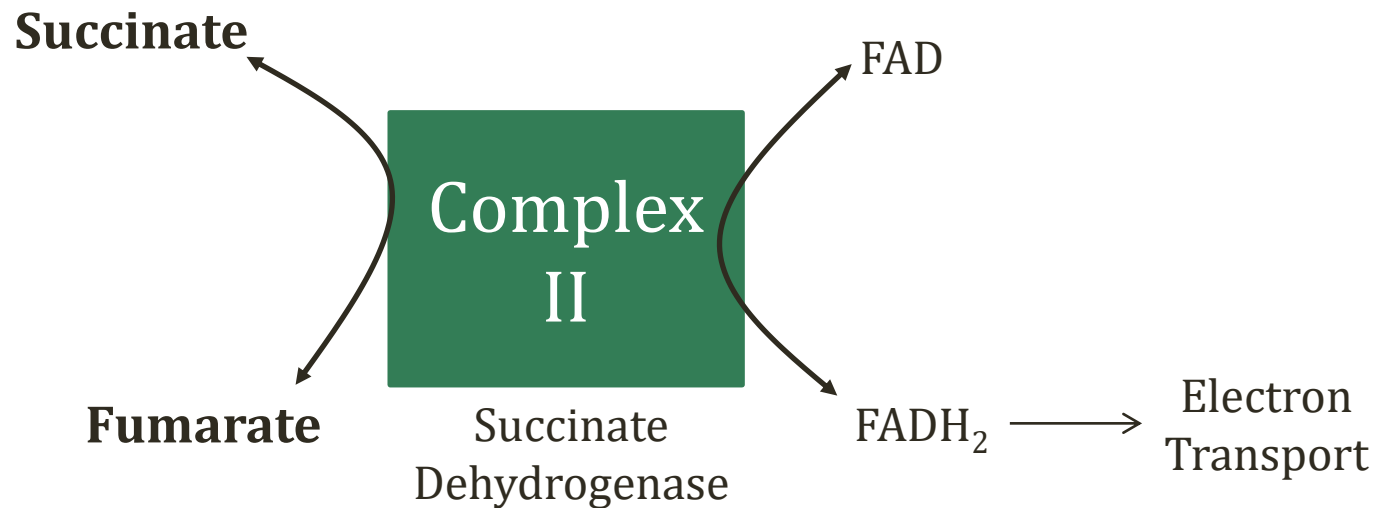
Succinate

- Succinyl-CoA synthase



Fumarate

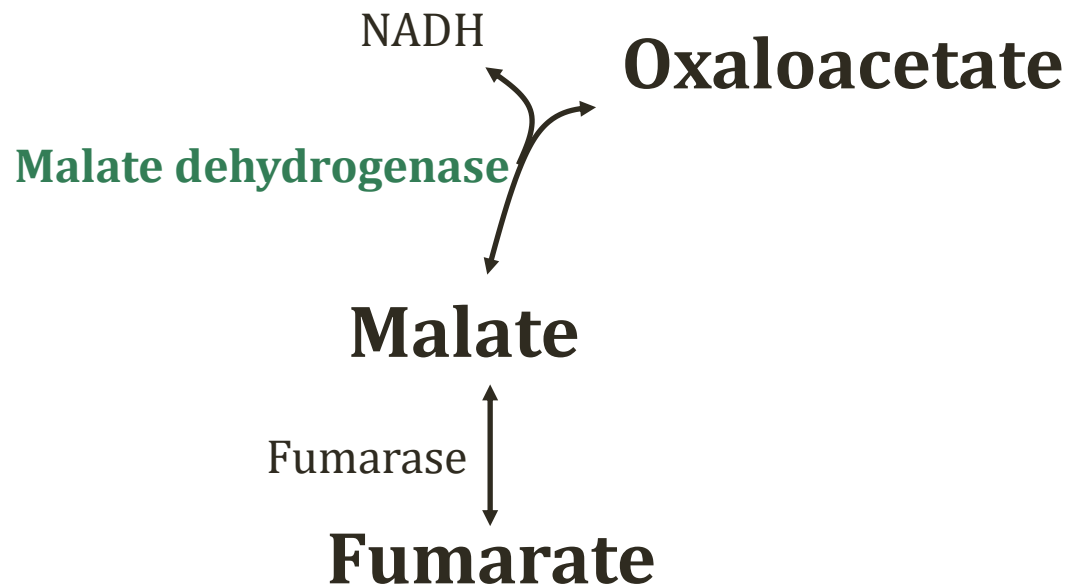
- Succinate dehydrogenase
- Unique enzyme: embedded mitochondrial membrane
- Functions as complex II electron transport



Fumarate

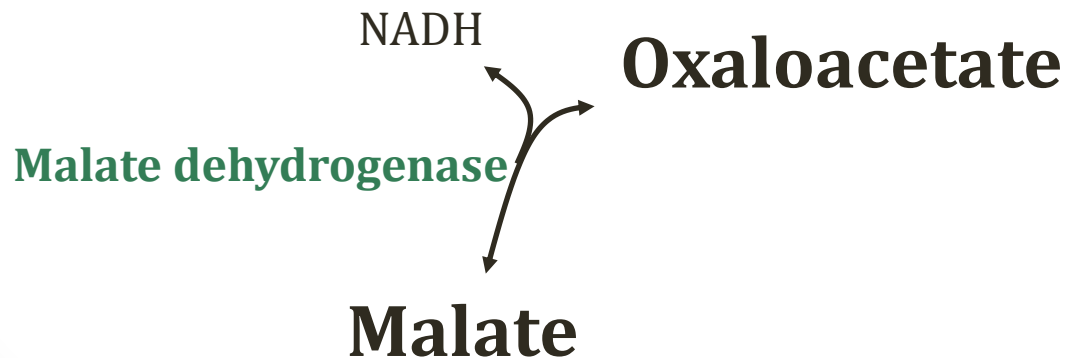
- Also produced several other pathways
 - Urea cycle
 - Purine synthesis (formation of IMP)
 - Amino acid breakdown: phenylalanine, tyrosine

Malate and Oxaloacetate



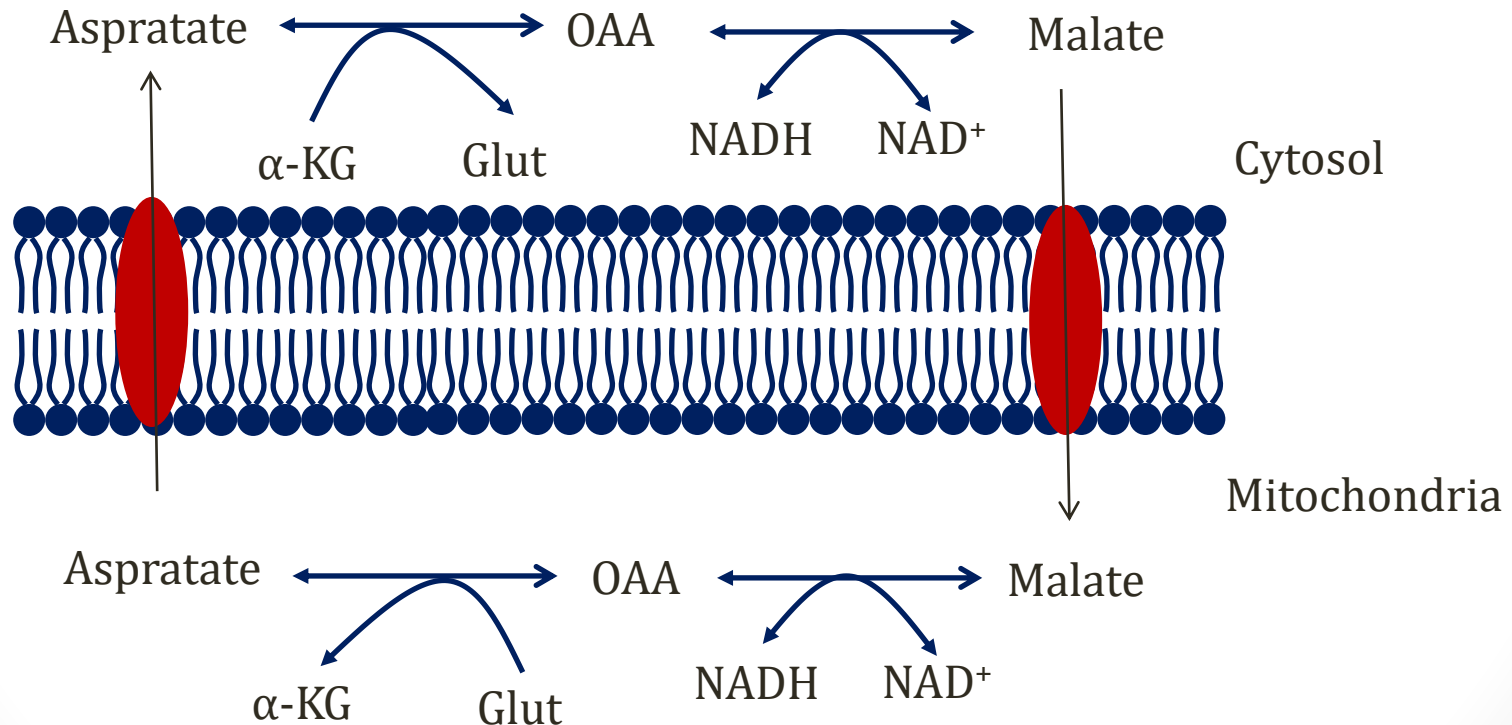
Malate Shuttle

- Malate “shuttles” molecules cytosol $\leftrightarrow$ mitochondria
- Key points:
 - Malate can cross mitochondrial membrane (transporter)
 - NADH and oxaloacetate cannot cross
- Two key uses:
 - Transfer of **NADH** into mitochondria
 - Transfer of **oxaloacetate** OUT of mitochondria



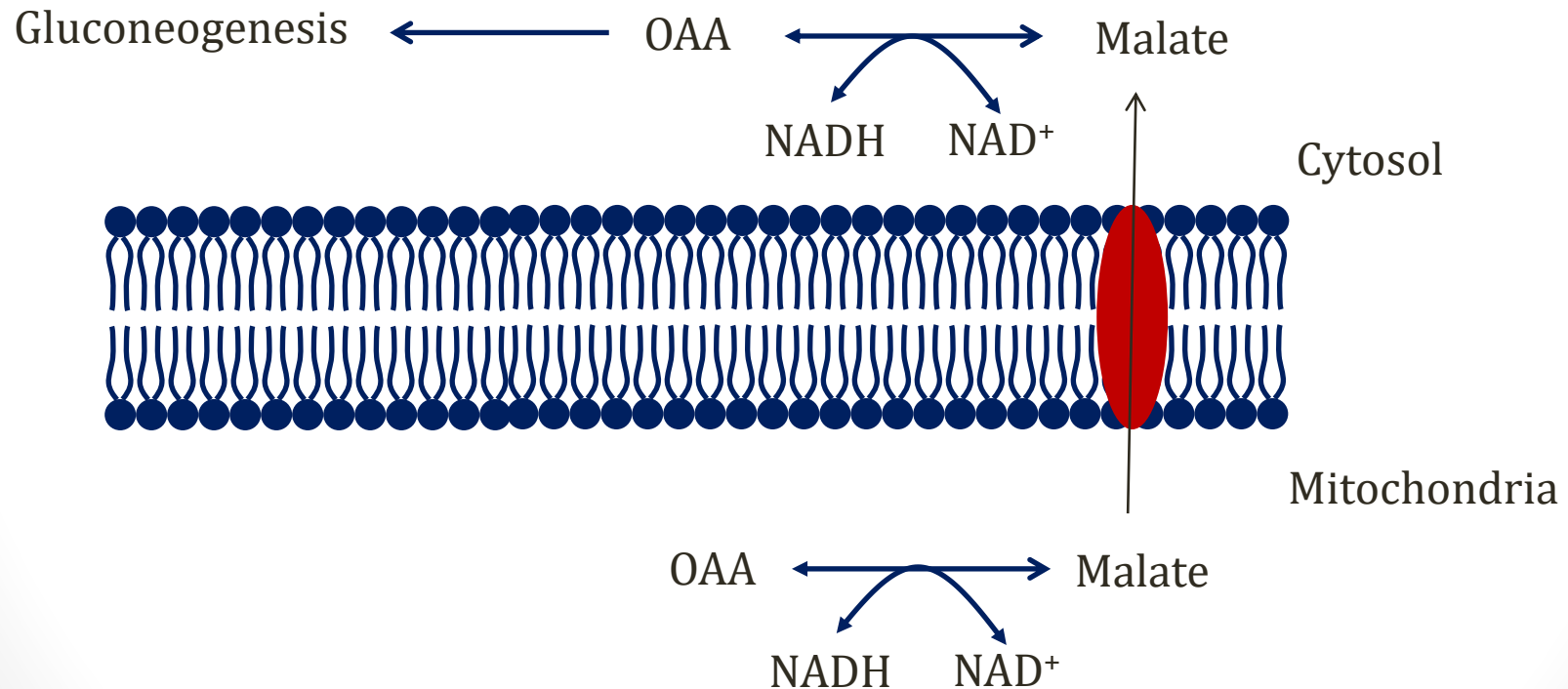
Malate Shuttle

- Use #1: Transfer of **NADH**

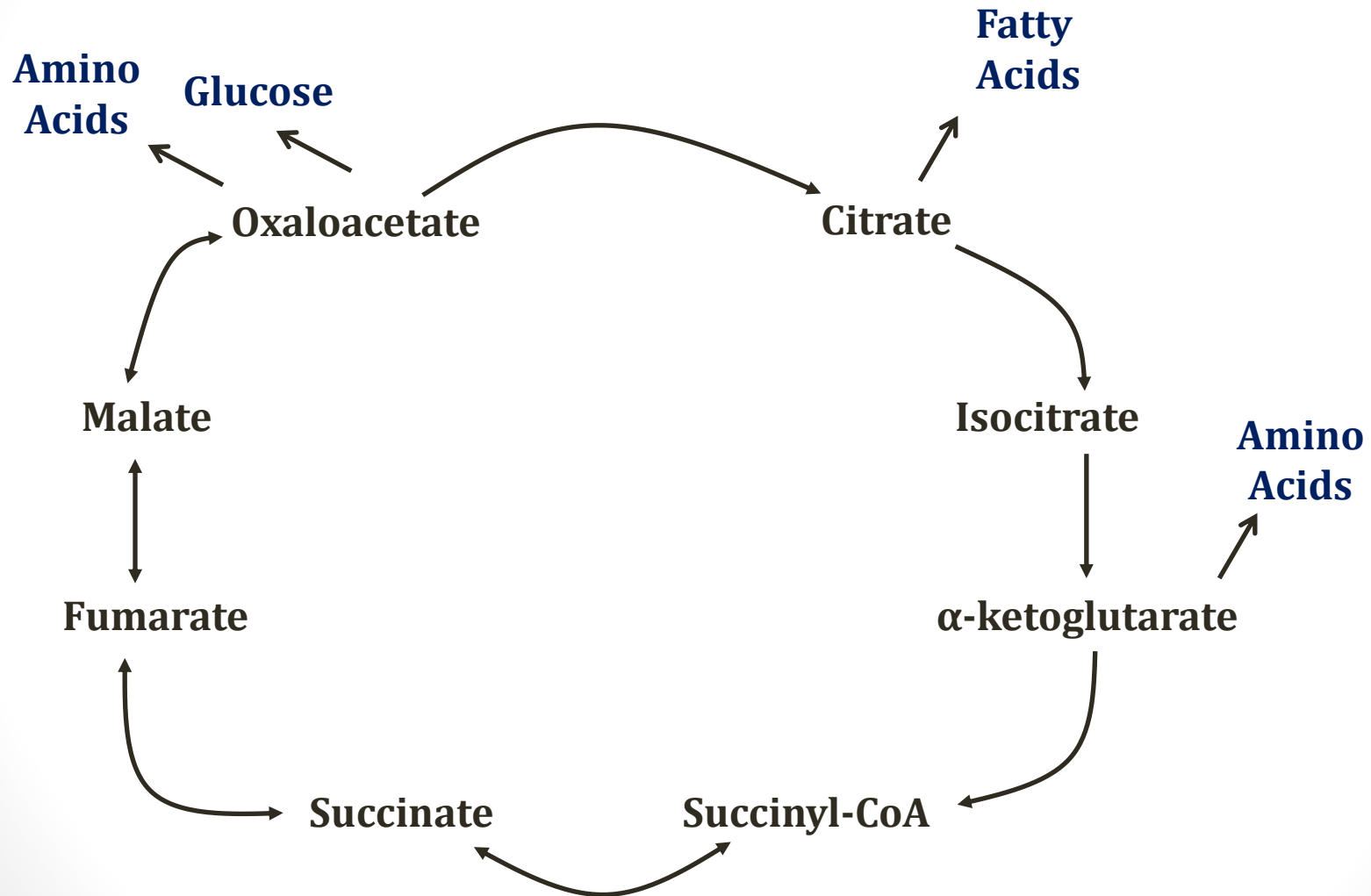


Malate Shuttle

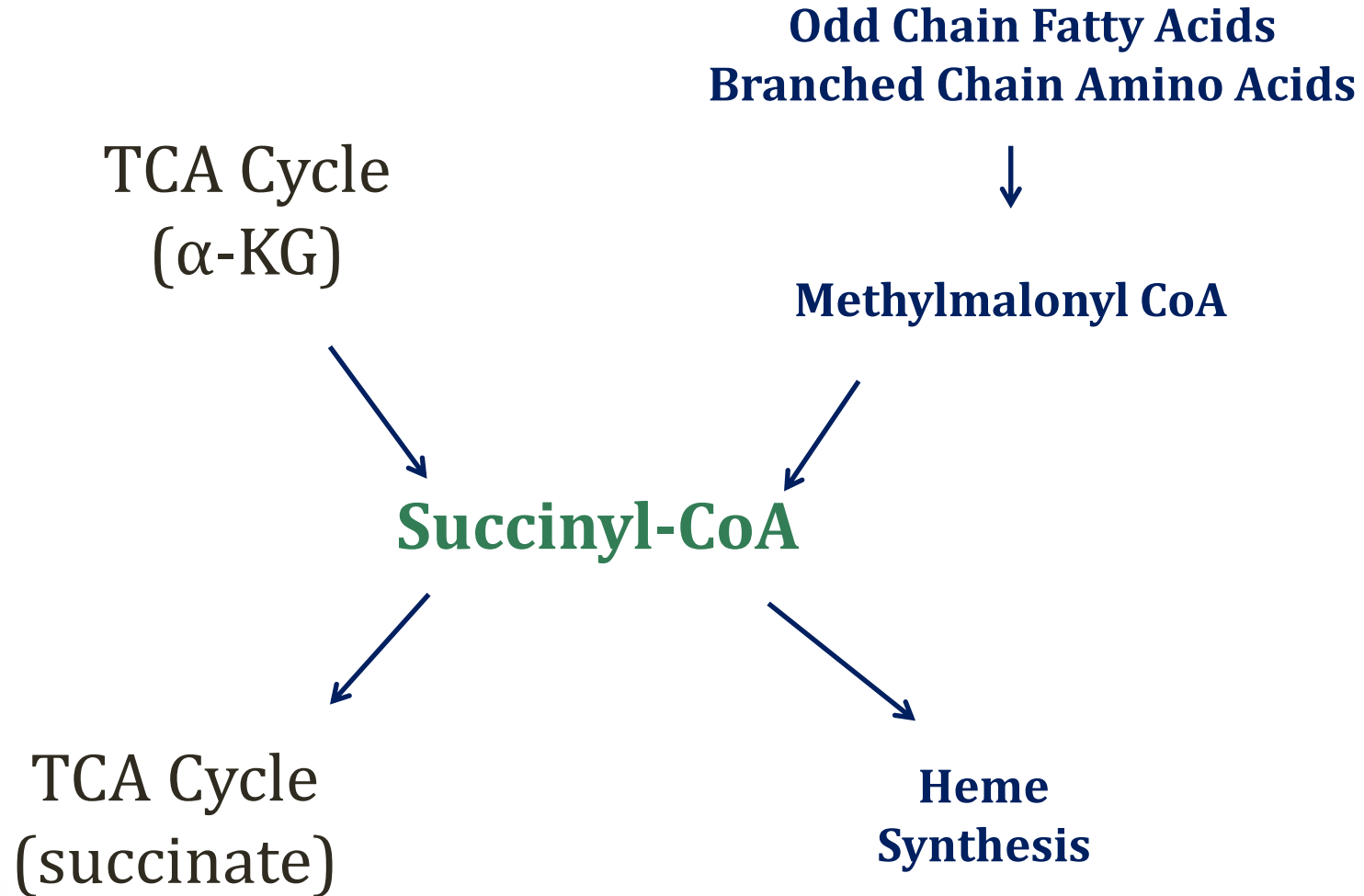
- Use #2: Transfer of oxaloacetate



TCA Intermediates



Succinyl CoA

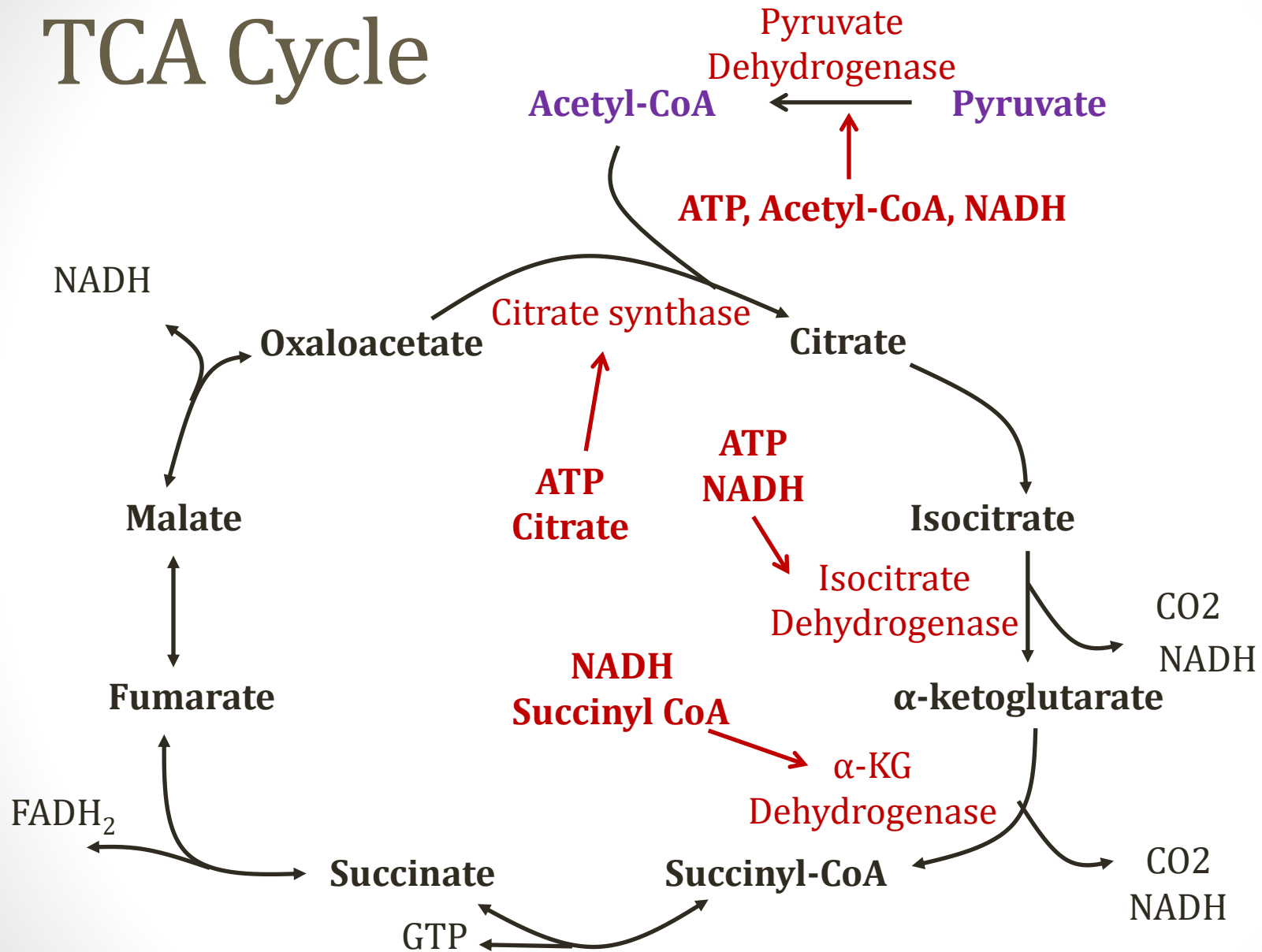


TCA Cycle

Key Points

- Inhibited by:
 - ATP
 - NADH
 - Acetyl CoA
 - Citrate
 - Succinyl CoA

TCA Cycle



TCA Cycle

Key Points

- Activated by:
 - ADP
 - Calcium

TCA Cycle

